D 143 | A loop is a measured hypothesis

PRIVATE FICTIONAL offline subscription lab. Rupees. No named-brand cohorts, product intervention, account access, participants, outreach or spend.

Finite cohort, not lifetime	100 initial paying subscribers. Month 1 is first paid service month. Active fractions at starts of months 1-12: 1, .80, .70, .65, .60, .55, .50, .45, .40, .35, .30, .25. All pay at month end; no within-month churn, refunds, reactivation or expansion. Revenue 150 and contribution fraction .70 per active user-month.
Cash timing and payback	CAC 260 per original customer at time 0. Twelve-month contribution 687.75, first three months 262.50. Discrete month-end payback by end month 3, excess 2.50. No fractional-month interpolation. Finite contribution proxy is not total lifetime profit. Support, fixed costs, taxes and other flows excluded, not verified absent.
Discounted value is separate	Editable monthly discount rate defaults 0 to preserve the case. PV discounts each end-month contribution by (1+r)^month, then deducts time-0 CAC once. Undiscounted payback is not discounted payback. Never mix annual and monthly rates or extrapolate the cohort beyond month 12.
Referral generations	100 initial users; .20 refer, 2 invites per referrer, .25 invite-to-paid conversion, no duplicate referrals. One generation gives 10 new, coefficient .1. If identically repeated for five generations, new counts 10, 1, .1, .01, .001 and cumulative incl. seed 111.111. Fractional counts are expectations, not actual people; no repeat referrals by earlier generations.
Different denominators	Separate MRR base 100, churn 20, contraction 5, expansion 30, reactivation 0: NRR 105%, GRR 75%. Separate activity sample has 80 active users and 460 active-days, mean 5.75 days, 12.5% at least 16 days. Separate original paid cohort retains 55/100 = 55%; activity distribution is not customer retention.
Evidence and action	Circular arrows, reviews and ad density do not prove a loop, actual brand CAC or retention. Activation-renewal correlation is not causal. Static prose is baseline, not a live report after edits. Actual evidence/permissions/owner/stop plan blank; no native cross-viewer, independent learner or overall 10 clearance.

Inputs | Four independent numeric families

Rupees, monthly rate and fractions. All synthetic. Cohort schedule separately editable; no reactivation allowed in that particular finite cohort.

Initial paid cohort, whole customers	100.00
Monthly revenue per active subscriber	150.00
Gross contribution fraction	0.70
CAC per original subscriber at time 0	260.00
Monthly discount rate, not annual	0.00

Referral seed users, whole count	100.00
Fraction referring per generation	0.20
Invites per referrer, expected count	2.00
Invite-to-paid conversion fraction	0.25

Inputs | Four independent numeric families

Rupees, monthly rate and fractions. All synthetic. Cohort schedule separately editable; no reactivation allowed in that particular finite cohort.

Separate starting original-cohort MRR	100.00
MRR lost to churn, nonnegative magnitude	20.00
MRR lost to contraction, nonnegative magnitude	5.00
MRR gained from expansion	30.00
MRR gained from reactivation	0.00

Separate original paid cohort size	100.00
Those original customers active now	55.00

Cohort economics gate (1=yes)	1.000000
Referral numeric gate (1=yes)	1.000000
MRR movement gate (1=yes)	1.000000
Customer count gate (1=yes)	1.000000

Gates test numeric consistency, not real economics, consent or causality. MRR reactivation belongs to original cohort, not new business. Starting MRR zero and customer cohort zero leave ratios undefined. Other families remain independent.

Cohort | Twelve month-end cash flows

Fractions of ORIGINAL cohort, no reactivation/expansion. Contribution per original subscriber, rupees. Inputs and static prose must be reviewed together after edits.

Month	Active fraction INPUT	Active customers	Revenue / original	Contribution / original	Cumulative contribution	PV contribution
1	1.00	100.000000	150.000000	105.000000	105.000000	105.000000
2	0.80	80.000000	120.000000	84.000000	189.000000	84.000000
3	0.70	70.000000	105.000000	73.500000	262.500000	73.500000
4	0.65	65.000000	97.500000	68.250000	330.750000	68.250000
5	0.60	60.000000	90.000000	63.000000	393.750000	63.000000
6	0.55	55.000000	82.500000	57.750000	451.500000	57.750000
7	0.50	50.000000	75.000000	52.500000	504.000000	52.500000
8	0.45	45.000000	67.500000	47.250000	551.250000	47.250000
9	0.40	40.000000	60.000000	42.000000	593.250000	42.000000
10	0.35	35.000000	52.500000	36.750000	630.000000	36.750000
11	0.30	30.000000	45.000000	31.500000	661.500000	31.500000
12	0.25	25.000000	37.500000	26.250000	687.750000	26.250000

Gate	1.000000

This no-reactivation case requires first month fraction 1 and nonincreasing fractions. Increasing fractions are invalid for THIS specification, not universally invalid customer retention: real reactivation can raise retention. Fractional active counts after edits are model expectations, not people observed.

Summary | Finite cash, not terminal LTV

Conditional authored arithmetic. Rupees except labeled counts, fractions or month indexes.

Sum of 12 active fractions	6.550000
12-month revenue per original customer	982.500000
12-month contribution per original customer	687.750000
First 3 months contribution per original customer	262.500000
12-month contribution minus time-0 CAC	427.750000
First 3 months contribution minus CAC	2.500000
Month-end undiscounted payback, 0=time 0	3.000000
PV of 12 end-month contributions	687.750000
Finite net PV after one time-0 CAC	427.750000
12-month contribution / CAC	2.645192

Baseline 6.55, 982.50, 687.75, 262.50, 427.75, 2.50, payback month 3, PV 687.75, net PV 427.75. If CAC is not recovered within 12 months, payback is undefined here, not never. CAC 0 gives time-0 payback but ratio undefined. No extrapolated lifetime, fractional-month payback or discounted payback calculated.

Loop | Five hypothetical new generations

Conditional authored arithmetic. Rupees except labeled counts, fractions or month indexes.

New-generation coefficient	0.100000
Seed users, generation 0	100.000000
Generation 1 NEW users	10.000000
Generation 2 NEW users	1.000000
Generation 3 NEW users	0.100000
Generation 4 NEW users	0.010000
Generation 5 NEW users	0.001000
All five generations NEW users	11.111000
Cumulative incl. seed after five generations	111.111000
Cumulative / seed	1.111110

Baseline coefficient .1; new 10, 1, .1, .01, .001; total new 11.111, cumulative 111.111. Each generation refers once with identical rates, no duplicates, no external acquisition and no earlier-generation repeat referrals. No cycle time, retained installed base, referral cost, market saturation or infinite-horizon claim. Coefficient above 1 is arithmetic, not proof sustainable growth.

MRR | Original revenue base only

Conditional authored arithmetic. Rupees except labeled counts, fractions or month indexes.

Gross retained MRR, before expansion/reactivation	75.000000
Net retained MRR incl. expansion/reactivation	105.000000
Gross revenue retention fraction	0.750000
Net revenue retention fraction	1.050000
Expansion plus reactivation MRR	30.000000

Baseline gross 75/net 105/GRR .75/NRR 1.05. No new-business revenue included. NRR above 1 does not mean all customers retained or profitable. Movements are aggregate nonnegative magnitudes with churn+contraction capped at opening MRR; this is a simplified convention, not reproduction of a provider event ledger. Actual status/past-due/reactivation and interval settings require reconciliation.

Engagement | Active users are not original subscribers

Separate fictional activity population. Thirty-day action frequency, exact supported days only; not a complete empirical product histogram or causal activation result.

Days active INPUT	Users at EXACT days INPUT
1.00	20.00
2.00	15.00
4.00	25.00
8.00	10.00
16.00	5.00
30.00	5.00

Histogram numeric gate	1.000000

Included active users	80.000000
Total active-user days	460.000000
Mean days among active users	5.750000
Share active at least 16 days	0.125000

Separate customer retention	0.550000

Baseline 80 active users, 460 active-days, mean 5.75, high-frequency share .125. Separate original customer retention .55. Do not divide these 80 by the other cohort 100: populations are not matched. A right-shift can reflect selection or survivor composition, not causal improvement. Natural weekly cadence does not require a daily smile.

Design plan | Specify before observing

No live intervention, research contact or participant test. Authored transfer prompts have no actual learner answers.

Loop measurement	Define seed event, deduplication, invitation exposure and conversion window, qualified new users, cost and cycle time. Map which output returns as a next-cycle input. Do not count earlier users as new each cycle or assume referrals are free. Circular diagram and referral button alone do not prove measured reinvestment.
Cohort boundary	Fix first paid interval, observation cutoff, original denominator and status rule. Separate activity/activation from paid customer retention and MRR. Unobserved future months are missing, not zero. This toy stipulates all 12 fractions, not censored actual data; do not extrapolate lifetime or infer competitors from reviews.
Cash and discounting	Collect at month ends; CAC at time 0. Apply contribution margin before payback; cash/profit differ if support, refunds, taxes or working capital are omitted. Specify periodic rate. Compare payback, finite net PV and risk; a faster payback alone does not prove a universally preferable investment.
Definitions and sources	ChartMogul cohort rules may include reactivated customers and past-due subscriptions until configured churn. Our finite cohort excludes reactivation, separate MRR model permits it. Reconcile interval and starting-base versus previous-interval views; provider benchmark correlations are not causal or consumer-app forecasts.
Activation versus causation	Predefine eligible users, activation event, outcome and follow-up. Selection, intent and survivorship can explain renewal correlation. Any randomized/nudge study would need actual authority, consent where required, assignment, missingness, uncertainty and economic guardrails. No live change or recruitment authorized.
Delayed transfer	Close notes: CAC 270 makes month-end payback 4, not 3; monthly discount .01 lowers PV but does not change undiscounted payback. Referral coefficient 1 yields five seed-sized new generations, not free infinite growth. Add original-cohort reactivation MRR 10: NRR 115%, GRR still 75%. Actual responses blank.

Actual plan | Evidence and approvals blank

No actual brand cohorts, participants, account access, experiment, contact, accepted owner or spending.

Question / population	Source / cutoff	Definition / timing	Observed vs assumed	Owner / acceptance	Permission / budget	Stop / delayed transfer

Record only real evidence if later authorized. Blank unknowns do not become zero churn, paid demand, accepted ownership or approved studies. No full source/AV/native Excel-Sheets parity/independent learner/account/field/legal/overall 10 clearance.

Evidence | Definitions and hypotheses, not brand results

Six URLs, three organizations. Practitioner frameworks and vendor documentation, not six independent effectiveness studies. Returned written content only.

Reforge loops	https://www.reforge.com/blog/growth-loops | Complete returned written July 31, 2018 article read. Reinvested output, system mapping and loop power; not diagram pixels, linked series/program, current company performance or guaranteed compounding. Its qualitative claim of defensibility is not independently tested here.
Chen / Li Jin	https://andrewchen.com/power-user-curve/ | Complete returned written essay read: L 30/L 7, activity action, cadence and population heterogeneity. No image plots inspected or historic named-product metric verified current. Retain author credit to Li Jin; not universal smile criterion or causal release analysis.
ChartMogul customers	https://help.chartmogul.com/article/159-cohort-customer-retention | Complete returned written page, updated October 22, 2026, read. Original customer cohort, Active/Cancelled/reactivation and interval views. No UI/account test, plot pixels, linked benchmark evidence or causal growth claim validation.
ChartMogul MRR	https://help.chartmogul.com/article/160-cohort-net-mrr-retention | Complete returned written page, updated October 22, 2026, read. Original MRR base, expansion/reactivation and past-due setting. Sentence saying above 100% may occur merely retaining all revenue is imprecise: unchanged 100/100 equals 100%, not above. Growth is needed for >100%.
ChartMogul gross/net	https://chartmogul.com/blog/retention-rate/ | Complete returned written February 19, 2021 article read. Gross excludes expansion, net includes original-customer expansion; customer and revenue retention differ. Images/linked benchmarks not read. Rhetorical retention-as-proof wording is not causal proof or universal health criterion.
Reforge monetization	https://www.reforge.com/guides/analyze-monetization-metrics | Complete returned written overview/reasons 1-3 read, including time value/payback cautions. Not complete 19-minute/four-step program, image-slide pixels or AV. Source says faster-payback company preferable; our recommendation remains conditional on costs, risk and value. No full book, independent learner or overall 10.

Failure log | Authored model checks only

Added after actual formula execution. Not observed product performance, learner mistakes, independent review or causal evidence.

1	PASS: 96 baseline formulas independently recomputed.
2	PASS: 24 changed sets plus restore.
3	PASS: 30 invalid inputs/2880 isolation checks.
4	PASS: 6 broken references/6 explicit checks.
5	PASS: independent cohort/loop/MRR/activity gates.
6	PASS: zero-base ratios and unrecovered payback undefined.
7	PASS: finite month-end payback; no fractional interpolation.
8	PASS: actual evidence/permissions/owner/stop plan blank.